

PAPER #90 MUGE Compressed Gravity: Newtonian Base + 9 Corrections

Title: MUGE Compressed Gravity: A 10-Term Framework Correcting Newtonian Gravity at Galaxy-to-Cosmological Scales

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Framework: MUGE (Multi-Unit Gravity Expression), UQFF Star-Magic

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Source Data: `validate_uqff_muge.py`, `source4.cpp` (10 Compressed functions), `compute_compressed_MUGE_SOURCE4`

Index Slot: 1.12 UQFF Master Calculators, Paper #90

Abstract

The MUGE Compressed gravity framework extends Newtonian gravity with 9 physics-motivated corrections spanning expansion, magnetic suppression, envelope, Ug-sum, cosmological constant, quantum, fluid, and dark matter perturbation effects. `validate_uqff_muge.py` validates the complete 10-term sum for 5 astrophysical systems (Sgr A*, M87, Sun, NeutronStar, Magnetar), confirming no NaN/Inf across 8 distinct radial ranges and 5 mass scales.

1. The 10-Term MUGE Compressed Framework

From `source4.cpp::compute_compressed_MUGE_SOURCE4`:

$$g_{\text{MUGE}}^{\text{Comp}}(r) = g_{\text{Newton}} + \sum_{k=1}^9 \delta_k(r)$$

Term Definitions

Term #	Symbol	Formula	Physics
0	<code>g_Newton</code>	GM/r	Base Newtonian gravity
1	<code>d_Expansion</code>	$H_0 r/6$	Hubble flow correction
2	<code>d_Super</code>	$-B/(4\pi r)$	Magnetic field suppression
3	<code>d_Envelope</code>	$g_N \cdot \eta_{\text{env}}/\eta_{\text{crit}}$	Gas envelope contribution
4	<code>d_Ug_sum</code>	$\sum U g_k/(r)$	UQFF Ug1+Ug2+Ug3+Ug4 sum
5	<code>d_Cosm</code>	$\Lambda c^2 r^3/3$	Cosmological constant (dark energy)
6	<code>d_Quantum</code>	$\hbar/(Mr^5)$	Quantum gravity correction
7	<code>d_Fluid</code>	$\eta v/(r)$	Navier-Stokes viscosity term
8	<code>d_Perturbation</code>	$d_{\text{DM}} g_N$	Dark matter perturbation

2. Term-by-Term Magnitudes at Sgr A* $r_{\text{horizon}} = 1.27 \times 10^6 \text{ m}$

Term	Value at r_horizon	Relative to g_N
g_Newton	$2.34 \times 10 \text{ m/s}$	1.000
d_Expansion	$7.8 \times 10^{?4} \text{ m/s}$	$3.3 \times 10^{?6}$
d_Super	$-1.2 \times 10^{?} \text{ m/s}$	$-5.1 \times 10^{?5}$
d_Envelope	$+8.5 \times 10^{-8} \text{ m/s}$	$+3.6 \times 10^{?}$
d_Ug_sum	$+1.4 \times 10^{-6} \text{ m/s}$	$+6.0 \times 10^{??}$
d_Cosm	$-6.5 \times 10^{?6} \text{ m/s}$	$-2.8 \times 10^{?8}$
d_Quantum	$+3.2 \times 10^{-47} \text{ m/s}$	$+1.4 \times 10^{?4?}$
d_Fluid	$+7.6 \times 10^{??} \text{ m/s}$	$+3.2 \times 10^{?}$
d_Perturbation	$+4.7 \times 10^{-4} \text{ m/s}$	$+2.0 \times 10^{-6}$
g_total	$2.340 \times 10 \text{ m/s}$	1.000002

No NaN/Inf - PASS. Total correction relative to Newton: < 5 ppm at r_horizon.

3. Dominant Corrections by Scale

Galactic Scale (r ~ kpc = $3 \times 10^5 \text{ m}$)

At galactic radius r = 1 kpc from Sgr A*:

Dominant corrections	Relative magnitude
d_DM Perturbation	$\sim 10^{?}$ (DM halo)
d_Expansion	$\sim 10^{?5}$ (sub-dominant at kpc)
d_Ug_sum	$\sim 10^{?7}$

? Dark matter perturbation dominates at galaxy scales. MUGE compressed reduces to DM+Newton.

Cosmological Scale (r ~ Gpc)

Dominant corrections	Relative magnitude
d_Expansion	$\sim 10^{?}$ (Hubble flow)
d_Cosm	$\sim 10^{?}$ (dark energy)

? Expansion and ? dominate at Gpc. MUGE compressed ? ?CDM-concordant.

4. Cross-System Validation (validate_uqff_muge.py)

All 5 systems from validator, all 10 MUGE terms verified finite:

System	M (kg)	r_test (m)	g_MUGE (m/s)	NaN/Inf
Sgr A*	8.0×10^6	1.27×10	234.3	None
M87*	1.26×10^4	1.95×10	2.21×10	None
Sun	1.99×10	6.96×10^8	274.3	None
NeutronStar	2.8×10	1.2×10^4	1.62×10	None
Magnetar	3.0×10	1.2×10^4	1.74×10	None

5. Relationship to UQFF_CompressedCalculator

The UQFF_CompressedCalculator (Paper #89) wraps the MUGE Compressed result and adds the d_Ug factor:

$$F_{\text{Comp}}^{\text{UQFF}}(r, t) = g_{\text{MUGE}}^{\text{Comp}}(r) + \delta_{\text{Ug}}(r, t)$$

Where d_Ug includes all 4 Ugk terms evaluated in the UQFF framework (not just their sum).

Summary

Scale	Dominant correction(s)	MUGE expansion
Near-horizon	d_Ug_sum + d_Perturbation	UQFF - GR
Galactic (kpc)	d_DM (0.1 g_N)	Dark matter-driven
Cosmological (Gpc)	d_Expansion + d_Cosm	?CDM-concordant
All scales	No NaN/Inf (5 systems)	Numerically stable

Source: `validate_uqff_muge.py` | `source4.cpp` compute_compressed_MUGE_SOURCE4 |
5 systems 10 terms all finite

*See
also:
PA-
PER_089
|
Part
of
the
Star-
Magic
UQFF
Whitepa-
per
Se-
ries.*

UQFF
com-
puted:
 MUGE
 buoy-
 ancy
 ratio
 U_{bi}/F_U
 =
 $[SSq] \cdot r/GM$
 =
 $5.7e-$
 $15.0e-$
 $4 =$
 $2.85e-$
 $4;$
 com-
 pressed
 MUGE
 base-
 line
 $g =$
 $5.4e-$
 7
 m/s
 at
 $r_{ISCO}.$

Appendix: UQFF Production Framework Reference (v4.75+)

Added by upgrade_early_whitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	UQFF exponential decay rate
$[SSq]$	0.57	Universal Quantized Factor
β_i	0.60-0.61	Buoyancy coupling coefficient
k_1	1.5	Ug1 DPM-dipole coupling
k_2	1.2	Ug2 outer-bubble charge coupling
k_3	1.8	Ug3 string-rotation coupling
k_4	2.0	Ug4 vacuum-concentration coupling
η	10^{-22}	Inertia tensor scale
$E_{\text{react}}(0)$	10^{46} J	Reference reactive energy

A.2 F_U Master Equation (Complete — 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 [\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}}]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
$-\Sigma\lambda_i\cdot U_i\cdot E_{\text{react}}$	4th dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420):

$\lambda_1=10^{-10}$, $\lambda_2=10^{-12}$, $\lambda_3=10^{-11}$, $\lambda_4=10^{-13}$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \times (1 + 10^{13} \Theta(\rho_{\text{SCm}} - \rho_c)) \times (1 + A_q \cos(\Delta\omega t))$$

Symbol	Value	Description
ρ_c	10^{15} kg/m^3	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\Delta\omega$	$2\pi/(434\cdot 365.25) \text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, ...)	Multi-scale field interactions
Buoyant	$\beta_i \times U_{bi}$	Expanding nebulae, stellar winds
Superconductive	$U_m \times (1 + 10^{13} \cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, Condensed-Physics.py, and CondensedPhysics2.py.